KPI Robustness in On-Time Performance Measurement for High-Frequency Bus Service

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Abstract. This study develops a reproducible robustness protocol for evaluating on-time performance (OTP) as a transit key performance indicator (KPI) computed from public General Transit Feed Specification (GTFS) and General Transit Feed Specification-Realtime (GTFS-RT) data. OTP is often reported as a single percentage. When it is computed from GTFS-RT trip updates, the analyst must make several methodological choices before the metric can be calculated. These choices include how repeated real-time predictions are matched and deduplicated, what lateness window defines "on time," how early arrivals are treated, and which stops and time periods are included. Using 21 days of self-archived Southeastern Pennsylvania Transportation Authority (SEPTA) GTFS-RT trip updates for high-frequency bus Routes 23 and 47, this study computes OTP across 1,080 reasonable specifications. Reported OTP ranges from 20.3% to 99.8% for Route 23 and from 20.3% to 99.3% for Route 47. The largest source of variation is the snapshot-selection rule used to decide which GTFS-RT update represents each stop event, accounting for a mean range of 30.7 percentage points across specifications—larger than any other dimension examined, including the well-studied lateness window. The study provides a transparent protocol for showing how sensitive a familiar KPI is to reasonable preprocessing and definition choices. The results suggest that GTFS-RT-based OTP reports should disclose both metric definitions and pipeline choices before presenting a single performance value.

1. Introduction

Reliability metrics are a central part of how public transit systems are evaluated. When these metrics are used to summarize and communicate system performance, they function as key performance indicators (KPIs). Agencies use them to monitor service quality, compare routes and operators, support internal decision-making, and communicate performance to the public. They also influence how outside observers judge whether service is improving or falling short. In that sense, a reliability KPI does more than describe the system. It helps shape what counts as good performance and what kinds of service problems become visible.

Among these measures, on-time performance (OTP) has a particularly strong practical role. It is easy to summarize, easy to publish, and easy to compare across routes or time periods. Because of that, OTP is often presented as if it were a simple and objective fact: a route is 82% on time, or it is not. But that appearance of simplicity can hide the number of decisions needed to produce the metric. What counts as "on time"? How much lateness is acceptable? Should early arrivals count as failures? Should OTP be measured at every stop, only at timepoints, or only at terminals? These are not small technical details. They affect what the metric means and how the resulting number should be interpreted.

This issue becomes even more important when OTP is computed from General Transit Feed Specification (GTFS) and General Transit Feed Specification-Realtime (GTFS-RT) data rather than from an internal agency performance system. Public GTFS and GTFS-RT feeds make transit data more accessible to researchers, journalists, app developers, and watchdog groups. At first glance, GTFS-RT-based OTP looks straightforward: match scheduled stop times with real-time updates, calculate delay, and classify each stop event as on time or not. In practice, however, GTFS-RT does not provide one clean historical arrival record for each stop event. Trip updates are live predictions, and repeated polling creates multiple overlapping records for the same scheduled stop. Before OTP can be calculated, an analyst must archive the feed, match updates to scheduled service, assign service dates, handle trips crossing midnight, filter the data, and decide which prediction represents each stop event.

The key argument is that those GTFS-RT pipeline choices are part of KPI methodology. They determine the stop-level delay values that are later classified as on time or not on time. A reported OTP value therefore reflects both metric-definition choices, such as lateness windows, and upstream data-processing choices, such as how repeated GTFS-RT predictions are deduplicated.

This issue is especially important for high-frequency bus service. On low-frequency routes, schedule adherence has a fairly direct passenger meaning because riders often time their arrival to a printed schedule. On high-frequency routes, passengers are more likely to arrive without planning around a specific trip. In those cases, headway regularity and waiting-time variability are often more meaningful from the rider's point of view than strict adherence to scheduled stop times. Even so, OTP remains widely used in practice, including on frequent routes where regularity may better capture the rider experience. That tension motivates this paper. Rather than replacing OTP with a headway-based metric, the study keeps OTP as the object of analysis and asks a more basic question: if OTP is going to remain a headline KPI, how much does it depend on the choices used to construct and compute it?

To answer that question, the study develops a reproducible robustness protocol for GTFS-RT-based OTP reporting. The protocol identifies reasonable preprocessing and definition choices, recomputes OTP across their combinations, and reports the resulting distribution rather than privileging one specification as the correct value. The approach draws on specification-curve analysis [Simonsohn et al., 2020] and multiverse analysis [Steegen et al., 2016] and adapts those ideas to descriptive KPI measurement rather than hypothesis testing.

In this setting, the purpose is to show how much a reported performance indicator changes when reasonable measurement choices change.

Empirically, the paper focuses on Southeastern Pennsylvania Transportation Authority (SEPTA) bus Routes 23 and 47. These routes were selected because they are the highest-frequency weekday pure bus routes in the SEPTA system and therefore provide useful cases for studying schedule-based OTP in a high-frequency setting. The analysis combines scheduled data from SEPTA GTFS static with self-archived GTFS-RT trip updates collected at 60-second intervals from April 4 through April 25, 2026. Real-time trip updates are matched to scheduled stop events, and OTP is calculated across 1,080 specifications that vary both metric-definition choices and GTFS-RT pipeline choices. The specification dimensions include snapshot selection, lateness window, treatment of early arrivals, stop selection, time of day, and delay basis.

The study addresses two research questions. RQ1 asks how much reported OTP changes when reasonable GTFS-RT preprocessing and definition choices vary. RQ2 asks which choices explain the largest share of that variation. These questions focus on the external-analyst setting and avoid treating public GTFS-RT estimates as SEPTA's internal OTP reporting. SEPTA's own reporting may rely on automatic vehicle location (AVL) data and agency-specific processing rules that are unavailable in the public GTFS-RT feed.

The results show that OTP is highly sensitive to these choices. For Route 23, reported OTP ranges from 20.3% to 99.8%, with a median of 67.0%. For Route 47, reported OTP ranges from 20.3% to 99.3%, with a median of 60.0%. The largest source of variation is the snapshot-selection rule used to decide which GTFS-RT prediction represents each scheduled stop event. At a fixed reference specification, changing only this rule moves OTP from 87.4% to 56.0% on Route 23 and from 86.2% to 48.8% on Route 47. This shows that GTFS-RT pipeline choices can change the interpretation of the final KPI.

The main contribution is therefore a reproducible robustness protocol for GTFS-RT-based OTP reporting. The protocol makes preprocessing and definition choices explicit, recomputes the KPI across reasonable alternatives, quantifies the spread, and identifies which choices drive the largest changes. The study also proposes a minimal disclosure checklist for GTFS-RT-based OTP reports. The aim is to make the measurement process behind OTP more transparent, rather than to introduce another reliability metric.

The rest of the paper is organized as follows. Section 2 reviews prior work on transit reliability metrics, inconsistencies in OTP and related measures, GTFS-RT-based performance analysis, and robustness-oriented methodological frameworks. Section 3 identifies the research gap and states the paper's contributions more directly. Section 4 presents the robustness protocol, data sources, stop-event reconstruction process, specification dimensions, and sensitivity analysis method. Section 5 reports the OTP specification curve, distribution of results, and sensitivity decomposition. Section 6 discusses the implications for GTFS-RT-based KPI transparency, proposes a minimal reporting checklist, and outlines limitations and future work.

2. Related Work

2.1. Transit Reliability Metrics and OTP

Transit reliability is a contested concept rather than a single fixed measure. Depending on the service pattern and the purpose of the analysis, it can be measured through schedule adherence, headway regularity, excess waiting time, travel time variability, or other stop- and passenger-centered indicators. This matters because different measures capture different kinds of service failure. In low-frequency settings, timetable adherence has a relatively direct passenger meaning because riders often coordinate their arrival to scheduled departures. In high-frequency settings, where passengers tend to arrive more randomly, headway regularity often becomes the more relevant reliability lens.

OTP occupies an awkward but important position in the measurement of high-frequency service. It remains widely used because it is easy to compute, communicate, and compare. However, schedule-based OTP may only partially represent the rider experience on frequent routes, where long gaps and bunching can matter more than strict adherence to a printed schedule. Li and Tan [2020], for example, develop a headway-based reliability framework using real-time data at network scale. Their work is especially relevant for high-frequency service because it focuses on irregularity in actual service delivery rather than only on adherence to scheduled stop times. For this study, that literature serves as an important comparison point. OTP remains the focus because it is still widely used despite those limitations, which makes its measurement choices worth examining more carefully.

The literature has also shown that reliability measurement is sensitive to definition. van Oort [2014] reports substantial inconsistency in how public transport authorities define and regulate service reliability and argues that traditional indicators can be misleading when they emphasize vehicle performance over passenger impacts. Godachevich and Tirachini [2021] make a similar point from another angle by showing that measured operator performance changes depending on which reliability index is used. Taken together, these studies suggest that reliability is defined through measurement choices, and those definitions affect evaluation.

This issue is especially important for OTP because OTP turns a continuous delay value into a binary label. A stop event is counted as on time or not on time according to a selected tolerance window, and that choice is rarely self-evident. Saberi et al. [2013] criticize conventional reliability measures because they often fail to capture the amount of delay, do not adequately distinguish between early and late departures, and rely on fixed tolerance windows. This critique is directly relevant here because this paper varies both lateness windows and early-arrival treatment as part of the robustness protocol.

Stop selection is another important source of variation.

Singh et al. [2024] show that the location and number of stops used to sample route-level OTP can affect whether the measure represents whole-route performance. Measuring only endpoints, for example, may not capture what happens along the route. This issue is included in the present study through specifications that use all stops, official timepoints, or terminals only.

Prior work therefore establishes that reliability metrics are methodologically shaped objects. They depend on what is measured, where it is measured, and how the metric is defined. The present study builds on that insight and extends it to the GTFS-RT pipeline itself. In addition to asking how OTP changes when the metric definition changes, it asks how OTP changes when the underlying stop events are reconstructed in different reasonable ways from public real-time feeds.

2.2. GTFS-RT and Event Reconstruction

The spread of GTFS static and GTFS-RT data has made transit performance analysis more accessible and, at least on the surface, more standardized. GTFS static provides planned service information such as routes, stops, trips, and schedules, while GTFS-RT provides live information such as vehicle positions, trip updates, and service alerts. This standardization matters because it allows analysts outside an agency to work with transit data in a more generalizable way than would be possible with agency-specific AVL systems.

Aemmer et al. [2022] show that GTFS-RT can support network-wide extraction of performance metrics and can be used to classify delays as systematic or stochastic. Their work demonstrates the value of GTFS-RT as a source for operational analysis, as well as for passenger-facing arrival predictions. At the same time, they also note that GTFS-RT performance analysis is still less developed than AVL-based analysis and that data quality and feed consistency remain challenges. GTFS-RT therefore creates an opportunity for public and generalizable performance analysis while still requiring careful processing.

Xian et al. [2024] make this processing problem especially clear. They describe GTFS-RT trip updates as forecasted arrival and departure times that are continuously updated. Because real-time data are generated at the time of each query, a dataset built from repeated polling contains redundant, duplicate, and outdated records. To extract actual arrival information, the protocol buffer messages must be processed into readable form, cleaned, and filtered. Their pipeline retains the most recent update for each vehicle-stop event, based on the assumption that the most recent update gives the best estimate of actual arrival or departure time.

That point is directly relevant to the present paper. Prior GTFS-RT work recognizes that real-time feeds require ingestion, cleaning, deduplication, and event selection. However, those processing choices are often treated as implementation steps needed to produce an analysis-ready dataset. This study treats them as methodological choices that can affect the final KPI. In particular, the rule used to select one retained GTFS-RT prediction for each scheduled stop event is substantive. It determines the delay value that is later classified as on time or not on time.

This distinction is important because GTFS-RT-based OTP can appear simpler than it is. A basic description might say that the analyst matches scheduled stop times to real-time trip updates, calculates delay, and reports the percentage of stops that are on time. But each part of that process contains choices: how to assign service dates, how to match trip updates to static schedules, how to handle repeated predictions for the same stop, and how to filter the resulting observations. A single-pipeline analysis hides those decisions once the final KPI is published.

The current study therefore extends GTFS-RT performance work in a specific way. It goes beyond asking whether a GTFS-RT pipeline can produce a plausible OTP estimate and asks how much that estimate changes when multiple reasonable pipeline and metric-definition choices are considered side by side. This is where the GTFS-RT literature connects to the robustness framework used in this paper.

2.3. Specification-Curve and Multiverse Analysis

The methodological inspiration for this paper comes from outside the transit literature. Simonsohn et al. [2020] argue that empirical results often depend on reasonable but non-unique analytic choices and propose specification-curve analysis as a way to report the full set of theoretically justified and statistically valid specifications. The strength of that approach is transparency: rather than treating one preferred specification as definitive, the analyst shows how results vary across the space of reasonable choices.

Steegen et al. [2016] develop a closely related argument through multiverse analysis. Their focus is on data processing and construction. The basic point is that raw data do not automatically become one uniquely correct analysis-ready dataset. They can often be filtered, coded, transformed, or aggregated in multiple reasonable ways, and those choices can change the final result. A multiverse analysis therefore evaluates the same question across a set of reasonably processed versions of the data to reveal which decisions matter most.

These ideas translate naturally to transit KPI construction. In this context, the relevant specifications are measurement rules rather than alternative regression models that remain reasonable within the domain: punctuality windows, treatment of early departures, stop-selection rules, temporal segmentation, delay basis, and procedures for selecting which GTFS-RT update becomes the retained stop event. The basic question is the same as in the reproducibility literature: does the reported result remain stable across the family of reasonable analytic paths, or does it depend heavily on a small number of choices?

This study uses the descriptive logic of specification-curve and multiverse analysis. It omits the joint statistical inference component of specification-curve analysis because OTP is a descriptive KPI rather than a hypothesis test. There is no null effect to test. The goal is instead to show the spread of plausible OTP values and identify which methodological

choices drive that spread.

3. Research Gap and Contributions

Prior work has established three important points. First, transit reliability is not a single, universally defined concept. It can be measured through schedule adherence, headway regularity, waiting-time-based measures, and other indicators, and the chosen indicator can affect how performance is interpreted. Godachevich and Tirachini [2021], for example, show that the measured performance of bus operators can differ depending on the reliability index used.

Second, OTP itself contains definitional choices. Saberi et al. [2013] criticize traditional OTP-style measures because they often count only whether a trip is late, do not adequately address early departures, and rely on fixed tolerance windows. Singh et al. [2024] focus on another part of the OTP problem: the number and location of stops sampled for route-level measurement. They note that operators vary widely in whether they measure at endpoints, all stops, or some subset of stops, and that this can bias route-level OTP.

Third, GTFS-RT has made public performance analysis more feasible, but it introduces its own data-processing requirements. GTFS-RT trip updates are live predictions rather than a finished historical event table. A historical dataset must be archived prospectively, parsed from protocol buffers, cleaned, matched to static schedules, and filtered. Repeated polling creates multiple predictions for the same stop event, so the analyst must decide which update to retain.

The less-developed part of this literature is the connection between these three points for OTP computed from the same underlying GTFS and GTFS-RT observations. The literature has examined reliability definitions, OTP stop sampling, headway-based alternatives, and GTFS-RT performance pipelines. However, GTFS-RT stop-event reconstruction has rarely been treated as part of the KPI methodology, and prior work has not quantified how much reported OTP changes when both pipeline choices and metric-definition choices vary together.

This study addresses that gap by developing and demonstrating a reproducible robustness protocol for GTFS-RT-based OTP reporting. The protocol identifies reasonable preprocessing and definition choices, recomputes OTP across their combinations, summarizes the resulting distribution, and identifies which choices drive the largest changes. Applied to SEPTA Routes 23 and 47, the protocol shows that the same underlying public data can support a wide range of reported OTP values. The largest source of variation is the snapshot-selection rule used to decide which GTFS-RT prediction represents each scheduled stop event.

The contribution is threefold. First, the study reframes GTFS-RT-based OTP as the output of a measurement pipeline rather than a value directly extracted from the feed. Second, it adapts specification-curve and multiverse logic to descriptive KPI reporting by showing the distribution of OTP values across reasonable specifications. Third, it proposes a minimal disclosure checklist for GTFS-RT-based OTP reports, emphasizing that analysts should disclose not only lateness windows and stop-selection rules, but also upstream pipeline choices such as service-date assignment, trip matching, and snapshot selection.

This contribution is methodological rather than metric-inventing. It shows how a familiar KPI can be reported more transparently when computed from public GTFS-RT data, without positioning the results as SEPTA's internal performance reporting.

4. Methodology

This section presents a reproducible robustness protocol for GTFS-RT-based on-time performance reporting and applies it to SEPTA Routes 23 and 47. The protocol makes the methodological choices behind OTP construction explicit and shows how much the reported KPI changes when those choices vary. The focus is the measurement problem faced by an external analyst using publicly available GTFS and GTFS-RT data, rather than SEPTA's internal OTP reporting, which may use internal AVL data and agency-specific processing rules.

In this setting, OTP is reconstructed from scheduled stop times and repeated real-time predictions. GTFS-RT trip updates provide a stream of predictions that must be archived, matched to static schedules, filtered, and reduced to one retained observation per stop event. Table 1 summarizes the protocol and how it is implemented in this study.

The rest of this section follows the protocol in order. It first describes the data and scope of the case study, then explains how stop events are reconstructed from GTFS-RT, how the specification dimensions are defined, and how OTP robustness is summarized.

4.1. Data and Scope

The empirical case study uses SEPTA bus Routes 23 and 47. These routes were selected because they are the highest-frequency weekday pure bus routes in the SEPTA system; here, "pure bus" distinguishes them from trolley, rail, or other non-bus services. They therefore provide a useful case for studying schedule-based OTP in a high-frequency setting, where OTP remains a common public-facing metric even though headway regularity may often be more meaningful from a rider perspective.

The study combines two public data sources. The first is SEPTA's GTFS static bus feed, which provides scheduled trips, stops, stop sequences, service calendars, and scheduled arrival and departure times. The second is SEPTA's GTFS-RT `trip_updates` feed, which provides real-time predictions for vehicle arrivals and departures. GTFS-RT is useful for external performance analysis because it is public and standardized, while it still requires reconstruction before it can serve as a historical record of realized stop events. Real-time feeds must be archived in advance, and repeated polling creates overlapping predictions for the same trip-stop event.

The GTFS-RT data were self-archived at 60-second inter-

Table 1: Robustness protocol for GTFS-RT-based OTP reporting.

Protocol step	Purpose	Implementation in this study
1. Define KPI and scope	State exactly what is being measured	Stop-event OTP for SEPTA Routes 23 and 47, weekdays, day-service trips
2. Archive and prepare data	Create a reproducible historical dataset	GTFS-RT `trip_updates` archived every 60 seconds and matched to SEPTA GTFS static schedules
3. Reconstruct stop events	Convert repeated real-time predictions into analyzable stop-level observations	Match GTFS-RT updates to scheduled route-trip-stop-service-date records
4. Identify reasonable choices	Make hidden methodology explicit	Snapshot rule, lateness window, early treatment, stop selection, time of day, delay basis
5. Recompute KPI across specifications	Avoid privileging one arbitrary specification	Fully crossed design producing 540 specifications per route
6. Summarize robustness	Show how much the KPI changes	Report minimum, quartiles, median, maximum, range, and standard deviation
7. Diagnose sensitivity	Identify which choices move the KPI most	Dimension-level range decomposition
8. Report required disclosures	Translate findings into transparency guidance	Minimal checklist for GTFS-RT-based OTP reporting

vals from April 4, 2026 through April 25, 2026. The archive contains 29,965 protobuf snapshots, which when parsed produce 402,682,884 stop-time-update records. Each record represents one prediction for one scheduled stop event in one snapshot; the same stop event therefore appears many times across the archive as the bus approaches and the prediction is updated. The analysis is filtered to weekday service on Routes 23 and 47, scheduled trips only, and day-service trips only. Owl trips are excluded because this paper focuses on high-frequency daytime service, while Owl service has different operating characteristics and lower frequency.

A rolling 4 AM Eastern service-date boundary is used instead of a midnight boundary. This is necessary because some late-evening day-service trips continue after midnight but are still scheduled as part of the previous service day in GTFS static data. For example, a trip that begins before midnight may have scheduled stop times encoded as 24:xx on the previous service date. Assigning these observations to the calendar date after midnight would break the match between real-time updates and the correct static schedule.

4.2. Reconstructing Stop Events from GTFS-RT

The pipeline converts raw GTFS-RT snapshots into matched stop-level delay observations. GTFS-RT protobuf snapshots are first parsed into tabular records containing trip, route, stop, stop sequence, predicted time, scheduled relationship, and snapshot timestamp. GTFS static tables are then loaded to define the scheduled service, including trips, stop times, calendars, and calendar exceptions.

A scheduled stop event is defined as a route, trip, service date, stop sequence, and stop ID from the static schedule. A real-time update is matched to that event when its trip and stop information align with the scheduled trip-stop record for the relevant service date. This creates an intermediate table in which a single scheduled stop event may have many GTFS-RT predictions, because the same stop can appear in multiple snapshots as the bus approaches it.

This repeated-prediction structure is the key methodological issue in the pipeline. GTFS-RT does not provide one final historical arrival time for each stop event in the same way an internal AVL archive might. Instead, an external analyst sees a stream of updates. To compute OTP, the analyst must decide which update from that stream represents the stop event. That decision is methodological because it determines the delay value that will later be classified as on time or not on time.

This study compares three snapshot-selection rules. The `closest` rule keeps the prediction whose predicted arrival or departure time is closest to the scheduled stop time. This can seem reasonable, but it biases results toward on-time predictions because OTP itself rewards closeness to scheduled time. The rule and the metric thus measure overlapping properties, producing selection bias on the very dimension being measured. The `last_before` rule keeps the latest snapshot whose snapshot timestamp occurs before the scheduled stop time. This approximates what a rider or app user would have seen immediately before the bus was scheduled to arrive. The `latest` rule keeps the chronologically last available prediction for each stop event. Because GTFS-RT predictions generally update as the bus approaches and passes the stop, this rule is treated in this study as the closest GTFS-RT-only proxy for realized stop behavior.

The difference among these rules is central to this paper. Lateness windows and early-arrival rules define how a delay value is labeled, but snapshot selection determines which delay value enters the calculation in the first place. In that sense, snapshot selection is upstream of the usual OTP-definition choices.

4.3. Specification Design

OTP is calculated across six specification dimensions. These dimensions were chosen because they represent reasonable

choices an external analyst might make when computing OTP from GTFS and GTFS-RT data. Some are familiar from prior OTP and reliability literature, such as lateness windows and stop selection. Others are specific to the GTFS-RT reconstruction pipeline, especially snapshot selection.

The lateness-window dimension tests three definitions of "on time": a strict 0 to +3 minute window, a moderate -1 to +5 minute window, and a wider -2 to +7 minute window. These bounds are treated as plausible alternatives rather than uniquely correct values. This is important because OTP converts continuous delay into a binary label, and prior work has criticized traditional OTP measures for relying on fixed tolerance windows and for not fully capturing the amount or direction of delay.

The early-treatment dimension tests whether early arrivals are counted as failures or ignored. This matters because early service can harm passengers differently from late service. A late bus increases waiting time, but an early bus may cause a passenger to miss the trip entirely.

The stop-selection dimension tests whether OTP is measured at all stops, only at official timepoints, or only at terminals. Prior work has shown that operators may sample different sets of stops and that endpoint-only measurement can misrepresent whole-route performance.

The time-of-day dimension separates AM peak, midday, PM peak, evening, and all-day service. The delay-basis dimension compares scheduled arrival time and scheduled departure time. In this SEPTA GTFS static feed, arrival and departure times are encoded identically at every stop, so this dimension produces no variation in the final results. It is still included because it represents a real methodological choice in GTFS-based analysis and because the lack of variation is itself informative about the structure of the public schedule data.

4.4. OTP Calculation and Robustness Analysis

For each matched stop event, delay is calculated as the difference between the selected GTFS-RT predicted time and the scheduled time from GTFS static:

$$\text{delay} = \text{predicted_time} - \text{scheduled_time}$$

A positive value means the vehicle is predicted to arrive or depart after the scheduled time. A negative value means it is predicted to arrive or depart early. OTP is then calculated as the percentage of included stop events that fall within the selected on-time window:

$$\text{OTP} = \frac{\text{number of on-time stop events}}{\text{number of included stop events}} \times 100$$

The six dimensions are fully crossed. There are three snapshot rules, three lateness windows, two early-treatment rules, three stop-selection rules, five time-of-day windows, and two delay bases. This creates 540 specifications per route:

$$3 \times 3 \times 2 \times 3 \times 5 \times 2 = 540$$

Because the study includes two routes, the final analysis contains 1,080 OTP estimates.

This design adapts the descriptive logic of specification-curve analysis and multiverse analysis to KPI measurement. It omits the joint statistical inference component from specification-curve analysis because OTP is a descriptive KPI rather than a hypothesis test.

After calculating OTP for all specifications, the analysis summarizes the distribution of OTP values for each route using minimum, quartiles, median, maximum, range, and standard deviation. It also decomposes sensitivity by specification dimension. For each dimension, the procedure holds all other choices fixed and measures the range of OTP produced by varying only that dimension. This decomposition is descriptive rather than causal in a statistical modeling sense, and it provides a transparent descriptive ranking of which methodological choices move the reported KPI the most.

4.5. Sample-Size Checks

Because some specifications restrict the data to smaller subsets, each specification is assigned a sample-size reliability tier. A sample-size threshold of 1,000 events was set as a robustness criterion. In this study, all 1,080 specifications exceeded this threshold; 216 fall in the 1,000-4,999 range and are flagged as lower-sample estimates, while 864 meet the 5,000-event tier and are treated as the primary reliable tier.

These tiers support transparency rather than selecting one correct OTP value. A specification based on fewer observations may still answer a meaningful question, such as terminal-only performance during a specific time period, but its value should be interpreted with more caution than an all-day, all-stop specification.

5. Results

Applying the robustness protocol produces 1,080 OTP estimates across two routes. This section reports the distribution of those estimates and identifies which methodological choices account for the largest changes. The results answer two questions. First, how much does reported OTP change when reasonable measurement choices vary? Second, which choices account for the largest changes? The main finding is that OTP varies by almost 80 percentage points across the specification set for both routes. Within that variation, the snapshot-selection rule is the largest single source of change, followed closely by the lateness window.

5.1. Distribution of OTP Across Specifications

Across the full set of specifications, reported OTP varies widely for both routes. For Route 23, OTP ranges from 20.3% to 99.8%, with a median of 67.0%. For Route 47, OTP ranges from 20.3% to 99.3%, with a median of 60.0%. This means that the same underlying GTFS and GTFS-RT observations can support OTP values that differ by 79.5 percentage points for Route 23 and 79.0 percentage points for Route 47.

Table 2: Specification dimensions used in the OTP robustness analysis.

Dimension	Variants	Purpose
Snapshot rule	`closest`, `last_before`, `latest`	Selects which GTFS-RT prediction represents each stop event
Lateness window	`0_3`, `neg1_5`, `neg2_7`	Defines the schedule-deviation range counted as on time
Early treatment	`count_early_as_late`, `ignore_early`	Determines whether early arrivals count as OTP failures
Stop selection	`all_stops`, `timepoints_only`, `terminals_only`	Determines which stops are included in route-level OTP
Time of day	`am_peak`, `midday`, `pm_peak`, `evening`, `all_day`	Tests whether OTP depends on the time window analyzed
Delay basis	`arrival`, `departure`	Calculates delay using scheduled arrival or departure time

Table 3: Distribution of OTP across reasonable specifications.

Route	Min	Q1	Median	Q3	Max	Range	SD
Route 23	20.3%	53.2%	67.0%	85.1%	99.8%	79.5 pp	20.7
Route 47	20.3%	47.6%	60.0%	80.9%	99.3%	79.0 pp	20.9

These values show why a single OTP number can be misleading without the methodology behind it. A report could describe the same route as performing poorly, moderately, or almost perfectly depending on how the GTFS-RT data are reconstructed and how "on time" is defined. The median values, 67.0% for Route 23 and 60.0% for Route 47, are useful summaries of the specification distribution, but they should not be interpreted as the one correct OTP value. The main result is the distribution itself.

Figure 1 shows the specification curve for all 1,080 OTP estimates. Each point represents one route-specific OTP estimate under one reasonable specification. The estimates are sorted from lowest to highest, and the lower panel identifies the methodological choices used for each point. The figure makes the main pattern visible: high OTP values are not randomly distributed across the specification set. They are associated with particular choices, especially the `closest` snapshot rule, wider lateness windows, and ignoring early arrivals. Lower OTP values are associated with stricter windows, counting early arrivals as failures, and using `last_before` or `latest` snapshot selection.

The curve should be read as a range of answers rather than as a search for one correct endpoint. Both ends are produced by reasonable choices, and the full curve shows what an external analyst could report after selecting only one plausible specification.

5.2. Sensitivity by Specification Dimension

To identify which methodological choices matter most, the analysis varies each dimension while holding the other dimensions fixed. This produces a range of OTP values for each dimension. A larger range means that changing that dimension has a larger effect on the reported KPI.

The snapshot-selection rule produces the largest average change in OTP, with a mean range of 30.7 percentage points. This means that, for a typical fixed combination of the other choices, changing only the snapshot rule shifts reported OTP by about 31 percentage points. The lateness window is the second-largest source of variation, with a mean range of 27.3 percentage points. These two dimensions dominate the results.

Table 4: Sensitivity decomposition by specification dimension.

Dimension	Mean range	Median range	Max range
Snapshot rule	30.7 pp	30.4 pp	50.1 pp
Lateness window	27.3 pp	28.8 pp	46.9 pp
Early treatment	14.8 pp	11.6 pp	48.6 pp
Stop selection	9.6 pp	8.3 pp	32.0 pp
Time of day	7.5 pp	4.1 pp	34.4 pp
Delay basis	0.0 pp	0.0 pp	0.0 pp

This ranking is important because the lateness window is an obvious methodological choice. Readers generally understand that a 0 to +3 minute window will produce a stricter OTP value than a -2 to +7 minute window. Snapshot selection is less visible. It happens before the OTP label is applied, during reconstruction of the stop event from repeated GTFS-RT predictions. Yet in this analysis, that upstream pipeline choice moves the final KPI more than any other dimension.

Early treatment also has a meaningful effect, with a mean range of 14.8 percentage points. Stop selection and time of day produce smaller average changes, but they are still large enough to affect interpretation in some cases. Delay basis produces no variation because SEPTA's GTFS static feed encodes scheduled arrival and departure times identically at every stop.

5.3. Snapshot Selection as the Largest Hidden Lever

The snapshot-selection result deserves special attention because it is both large and easy to miss. GTFS-RT trip updates are repeated predictions rather than a historical table of final arrivals. A bus stop can appear in many snapshots, and the analyst must decide which prediction to keep. That choice

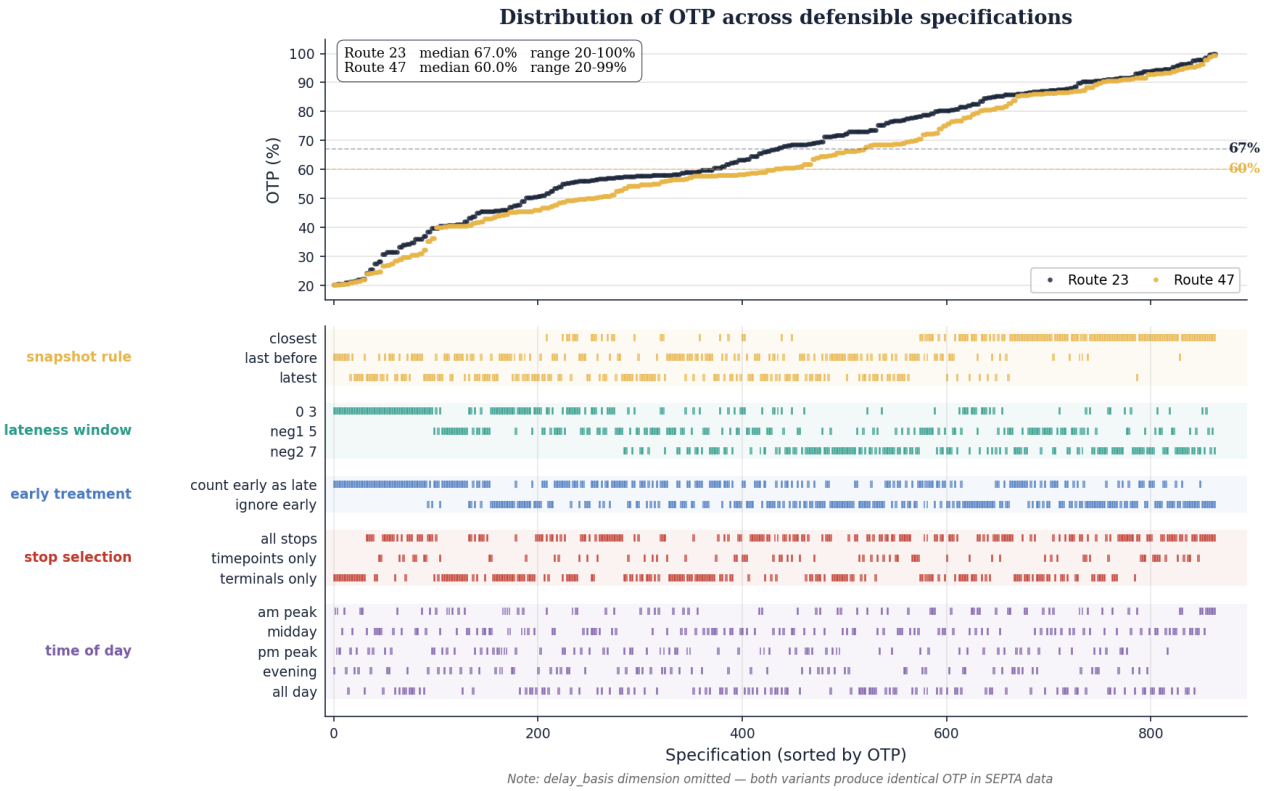

Figure 1: Specification curve for OTP across all specifications. The upper panel plots OTP values sorted from lowest to highest. The lower panel shows the specification choices behind each estimate, including snapshot rule, lateness window, early treatment, stop selection, time of day, and delay basis.

Table 5: OTP by snapshot rule at fixed reference specification.

Route	Closest	Last before	Latest
Route 23	87.4%	60.7%	56.0%
Route 47	86.2%	55.9%	48.8%

determines the delay value that is later classified as on time or not on time.

The effect is clear at a fixed reference specification. Holding the lateness window at -1 to +5 minutes, counting early arrivals as failures, using all stops, using the all-day time window, and calculating delay from scheduled arrival time, the three snapshot rules produce very different OTP values.

For Route 23, the `closest` rule produces an OTP value of 87.4%, while `latest` produces 56.0%. For Route 47, `closest` produces 86.2%, while `latest` produces 48.8%. In both cases, the `closest` rule produces OTP values roughly 30 percentage points higher than the `latest` rule.

The reason is straightforward once the rule is examined. The `closest` rule selects the prediction whose predicted time is closest to the scheduled time. Since OTP is also based on closeness to the scheduled time, this rule tends to retain predictions that already look more on time. It therefore builds an on-time bias into the event-reconstruction step.

The `latest` rule answers a different question. It keeps the final available GTFS-RT prediction for a stop event. Because real-time predictions usually update as the bus approaches or passes the stop, this rule is treated in this study as the closest available GTFS-RT-only proxy for realized stop behavior. Although it is not internal AVL ground truth, it avoids selecting predictions because they happen to be near the schedule.

The `last_before` rule falls between these two interpretations. It captures the last prediction available before the scheduled stop time, which is useful for approximating what riders or applications might have seen shortly before the bus was supposed to arrive. Its OTP values are lower than `closest` but usually higher than or near `latest`.

The broader implication is that snapshot selection is more than a technical implementation detail. It is part of the KPI methodology. An OTP report based on GTFS-RT should disclose which snapshot-selection rule was used, because that choice can change the reported result by an amount large enough to alter the interpretation of service reliability.

5.4. Other Specification Effects

The lateness window is the second-largest source of variation. This is expected because changing the on-time window directly changes the definition of success. A strict 0 to +3 minute window gives buses little room for either early or late deviation. A wider -2 to +7 minute window classifies more observations

as on time. The sensitivity decomposition shows that this choice changes OTP by 27.3 percentage points on average.

Early treatment is also important. Counting early arrivals as failures produces lower OTP values than ignoring early arrivals. This matters because early service can be harmful from a passenger perspective. A late bus increases waiting time, but an early bus may cause a rider to miss the trip completely. The results confirm that the decision to count or ignore early arrivals can meaningfully affect the final KPI.

Stop selection has a smaller average effect than expected. Prior work, particularly Singh et al. [2024], suggests that measuring only certain stops, especially endpoints, can bias route-level OTP. This concern is still valid, and the maximum stop-selection range in this study reaches 32.0 percentage points. However, the mean range is 9.6 percentage points, which is smaller than the effect of snapshot rule, lateness window, or early treatment. One possible explanation is that the GTFS-RT reconstruction step introduces variation that scales with the number of opportunities for prediction drift, while the all-stops, timepoints, and terminals subsets all draw from a relatively consistent population of well-predicted stops. The result is a less consequential dimension than the prior literature would suggest, and warrants further investigation across additional agencies and feed structures. In this case study, where the analyst reconstructs stop events from GTFS-RT, upstream snapshot selection matters more than the choice between all stops, timepoints, and terminals.

Time of day also affects OTP, but less consistently. The mean range is 7.5 percentage points, while the maximum range is 34.4 percentage points. This suggests that time-of-day effects can be important in specific combinations of choices, but they are not the main driver of variation across the full specification set.

Delay basis produces no variation. This is because SEPTA's static GTFS data encode the same scheduled time for arrival and departure at every stop. As a result, calculating delay from scheduled arrival time or scheduled departure time produces identical OTP values. The dimension remains conceptually relevant. In other GTFS feeds, arrival and departure times may differ, especially at timepoints or stops with scheduled holding. In this case, the zero effect is a transparency finding about the public schedule data itself.

5.5. Why the Extremes Are Not Anomalies

The highest and lowest OTP values in the specification curve should not be treated as data errors. They are produced by combinations of individually reasonable choices.

The highest OTP specifications share several features. They use the `closest` snapshot rule, ignore early arrivals, apply a lenient lateness window, and concentrate in the AM peak period. The combination is consequential: morning buses tend to run early in light traffic, and pairing this with the `ignore_early` rule classifies those early arrivals as on time. Each choice can be defended in isolation, but their interaction selects strongly for predictions that look on time, producing OTP values near 99%.

The lowest OTP specifications also come from reasonable choices. They generally use `last_before` or `latest` snapshot selection, apply the strict 0 to +3 minute window, count early arrivals as failures, and focus on terminals. This combination creates a stricter definition of on-time performance and uses snapshot rules closer to realized stop behavior. The result is OTP values near 20%.

This is exactly why the specification-curve approach is useful. The extremes should not be dismissed simply because they are far apart. They show what happens when reasonable choices are combined in different ways. Neither extreme should be treated as the correct value by itself. Instead, the spread between them shows how much interpretation is hidden when OTP is reported as a single number.

These findings support the central argument of the paper. When OTP is computed from public GTFS-RT data, the reported value depends not only on the definition of "on time," but also on how the analyst reconstructs stop events from repeated real-time predictions. A single OTP value is therefore incomplete unless the methodological choices behind it are disclosed.

6. Discussion and Conclusion

The results show that GTFS-RT-based OTP is highly sensitive to reasonable methodological choices. Across SEPTA Routes 23 and 47, reported OTP spans almost 80 percentage points depending on how stop events are reconstructed and how the on-time label is defined. This should be read as a transparency result rather than as a claim that OTP is useless or that any single specification is intentionally misleading. OTP is the output of a measurement process, and that process needs to be visible.

6.1. Interpreting the Specification Spread

The most important result is the size of the specification spread. Across both routes, reported OTP varies by nearly 80 percentage points. This is too large to treat as minor technical variation. A single reported OTP value could support very different interpretations of the same service depending on which specification produced it.

Different specifications answer different questions, so the values in the range should be interpreted in context. A strict window that counts early arrivals as failures asks whether buses closely follow the schedule. A wider window that ignores early arrivals asks a more lenient question about whether buses are generally close enough to schedule. A terminal-only specification focuses on endpoint performance, while an all-stops specification captures the full route more directly. These choices reflect different interpretations of what OTP should represent.

This is why the specification-curve framework is useful. It shifts the focus away from choosing one specification and treating it as the truth. Instead, it shows the range of results produced by reasonable choices and identifies which choices

drive the biggest differences. In this study, the distribution itself is the main finding. The median OTP values, 67.0% for Route 23 and 60.0% for Route 47, are useful summaries, but they should not replace the full robustness view.

6.2. GTFS-RT Pipeline Choices as KPI Methodology

The central conceptual finding is that GTFS-RT pipeline choices are part of KPI methodology. Prior reliability literature already shows that metric definitions matter. Lateness windows, early-arrival treatment, stop selection, and the choice between schedule-based and headway-based reliability all affect how service quality is measured. This study confirms those concerns and extends them upstream.

When OTP is computed from GTFS-RT `trip_updates`, the analyst must first reconstruct stop events from a stream of repeated predictions. This reconstruction step happens before the usual OTP definition is applied. The analyst has to decide which real-time update represents the stop event. That choice determines the delay value that is later classified as on time or not on time.

The results show that this upstream choice is substantial. Snapshot selection produces a mean OTP range of 30.7 percentage points, the largest of any specification dimension in the analysis. At the fixed reference specification, the `closest` rule produces OTP values of 87.4% for Route 23 and 86.2% for Route 47, while the `latest` rule produces 56.0% and 48.8%. In both cases, changing only the snapshot-selection rule changes the interpretation of performance.

This happens because the `closest` rule selects the prediction closest to the scheduled time. Since OTP also rewards closeness to the schedule, this rule tends to retain predictions that already look more on time. The `latest` rule, by contrast, keeps the final available prediction for the stop event and is therefore treated as the closest GTFS-RT-only proxy for realized stop behavior in this study. This distinction leaves room for different use cases: `latest` is not perfect ground truth, and `closest` may still be useful in some contexts. Still, the snapshot-selection rule must be disclosed because it strongly affects the final KPI.

This finding matters especially for external analysts. Agencies may have access to internal AVL records, quality-control procedures, and official performance-reporting rules. Outside researchers, journalists, app developers, and watchdog groups often do not. They may rely on public GTFS-RT feeds because those feeds are standardized and accessible. For those users, pipeline choices define the measurement.

6.3. A Minimal Disclosure Checklist

The results suggest that GTFS-RT-based OTP reports should include more than the final percentage. A reported OTP value is only interpretable if the reader knows how the stop events were reconstructed and how the on-time label was defined. Based on this case study, the most important disclosure is the snapshot-selection rule, because it produced the largest average change in OTP. However, the other specification dimensions should also be reported because they define the scope and meaning of the metric.

The case study is limited to two routes, one agency, and one GTFS-RT feed, so the checklist should be understood as a practical reporting aid rather than a universal industry standard. The analysis still supports a minimal disclosure checklist for external GTFS-RT-based OTP reporting. The checklist in Table 6 translates the robustness protocol into a practical reporting artifact. Its purpose is to make those choices visible, whatever choices an analyst makes.

The snapshot-selection item is especially important because it is easy to overlook. Lateness windows and stop-selection rules are usually recognizable as metric-definition choices. Snapshot selection happens earlier, during the reconstruction of stop events from repeated GTFS-RT predictions. Yet in this study, that upstream pipeline choice changed reported OTP more than any other dimension. For that reason, a GTFS-RT-based OTP report that omits its snapshot-selection rule leaves out one of the most important parts of the measurement process.

6.4. Limitations

This study has several limitations. First, it uses one agency and two routes. SEPTA Routes 23 and 47 are useful cases because they are high-frequency weekday bus routes, but the exact numerical results may vary across agencies, routes, or feed structures. A different GTFS-RT feed may have different update behavior, prediction quality, stop-time conventions, or data gaps.

Second, the archive covers a limited time period. The study uses data collected from April 4 through April 25, 2026 and filters the analysis to weekday day-service observations. This provides enough data to demonstrate the robustness protocol, while offering limited support for broad claims about SEPTA's long-term reliability.

Third, this study lacks internal AVL data. The `latest` snapshot rule is treated as the closest GTFS-RT-only proxy for realized stop behavior, and remains a proxy. Internal AVL data could provide a stronger ground-truth comparison and would allow future work to evaluate which GTFS-RT reconstruction rule most closely approximates actual arrivals.

Fourth, the specification set is broad but incomplete. It includes six reasonable dimensions, and other choices could also matter. For example, future work could vary trip-matching rules, missing-data thresholds, route-direction handling, alternative time-of-day windows, or different approaches to canceled and unscheduled trips. The protocol is designed to make such additions possible.

Fifth, the analysis is restricted by design to weekday day-service. Owl service is excluded, and weekend service for Routes 23 and 47, which the pipeline did capture, was not analyzed. These were treated as scope choices rather than specification dimensions because the paper focuses on high-frequency weekday service. A fuller robustness protocol could include weekday/weekend and day-service/Owl as additional dimensions, which would test whether the snapshot-selection

Table 6: Minimal disclosure checklist for GTFS-RT-based OTP reporting.

Reporting item	Why it matters
GTFS-RT feed type and archive interval	Determines what real-time observations are available
Static GTFS version or date	Defines the schedule used as the benchmark
Service-date boundary	Affects trips that cross midnight
Route, trip, and stop matching rule	Determines which real-time updates are assigned to which scheduled events
Snapshot-selection or deduplication rule	Determines which prediction represents each stop event
Lateness window	Defines what counts as on time
Early-arrival treatment	Determines whether early service is counted as a failure
Stop-selection rule	Determines whether OTP reflects all stops, timepoints, or terminals
Time period and service filters	Defines the scope of the reported KPI
Sample size	Shows how many observations support the reported value
Excluded trips or records	Documents filtering choices such as Owl service or non-scheduled trips

finding generalizes to lower-frequency service patterns.

Finally, the focus on OTP is itself a limitation. For high-frequency bus service, headway regularity and waiting-time variability may better represent rider experience than schedule adherence. The study does not position OTP as the main reliability metric for high-frequency service. It starts from the practical fact that OTP remains widely used and asks how transparently it can be reported when computed from public GTFS-RT data.

6.5. Future Work

Future research can extend this work in several directions. The most direct extension is to apply the robustness protocol to more routes, agencies, and feed types. A multi-agency comparison would show whether snapshot selection is consistently important or whether its effect depends on local GTFS-RT update behavior.

A second extension is to compare GTFS-RT-derived stop events with internal AVL or other ground-truth arrival records. That would make it possible to evaluate not only how much snapshot-selection rules change OTP, but also which rule most accurately approximates realized service.

A third extension is to apply the same robustness approach to headway-based reliability. This would be especially useful for high-frequency service, where riders may care more about even spacing between buses than strict adherence to a printed schedule. A robustness protocol for headway adherence could test choices such as headway tolerance, stop selection, bunching definitions, and time-of-day windows.

A final extension would be a simple dashboard or reporting interface for KPI robustness. Instead of showing only one OTP value, a dashboard could let users see the full distribution of reasonable values and filter by specification choices. This would make the methodological uncertainty easier to communicate to nontechnical audiences.

6.6. Conclusion

This study developed and demonstrated a reproducible robustness protocol for GTFS-RT-based OTP reporting. Applied to SEPTA Routes 23 and 47, the protocol showed that reported OTP varies by approximately 80 percentage points across reasonable specifications. The largest source of variation was snapshot selection, an upstream GTFS-RT pipeline choice that determines which prediction represents each scheduled stop event.

The main conclusion is about reporting rather than abandonment: OTP should not be reported as a single unqualified number when it is computed from public GTFS-RT data. A transparent report should disclose how stop events were reconstructed, how on-time status was defined, and how sensitive the result is to reasonable alternatives. That transparency makes the KPI more interpretable and gives readers a clearer understanding of what the reported number actually means.

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